

Do the actual terminal droplets justify changing the top?

Not yet established. Available Kühni experiments establish real terminal *active-stage* distributions, but do not determine whether droplets below 0.500, 1.000, 1.455 or 2.081 mm materially contribute to this NMP withdrawal load. Neither negligible fines nor a mandatory coalescer follows.

Disposition: retain the Ø700 × 1600 mm top spatial reservation [I; historically E]. Bare Ø700 remains the baseline *candidate*, not a separation PASS. No enlargement, internal collector or external guard is justified for selection now. All active geometry and the complete bottom arrangement remain fixed.

Correction to the previous selection interpretation

This investigation **supersedes the selection interpretation** in *top-disengager-final-review* that treated an assigned 500 µm screen as a design capture basis or gave an aided/guard route preferred status. **That duty and preselection are withdrawn here.** The conditional single-drop arithmetic is not erased, but its failure cannot establish equipment need. Earlier reports remain unmodified for audit.

“Material” requires the physical dispersed load, transported size distribution and an agreed carryover specification. None is established by the calculated active-section mean. A small volume fraction may still matter to a stringent specification; a large local count fraction is not automatically a large solvent loss.

Classification used throughout	Meaning
I · Inherited	Existing basis or cited donor measurement; not newly measured or approved.
C · Calculated	Mathematical/model result conditional on inputs.
A · Assumed	Unverified distribution, transferred width or physical model.
E · Engineer-selected	Reporting cutoff, comparison diameter or legacy screening factor; not a measured duty.
HOLD	Unresolved service evidence or equipment performance.
PASS	Only the explicitly named numerical or preservation check; never engineering acceptance.

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Scientific answer in one sentence

Use genuine Kühni DSD evidence to frame a **testable family and transport question**, not to manufacture a fine-capture duty from d32 or to choose a large vessel.

What the terminal-stage literature really supports

S1 · Oliveira et al. (2008): full primary evidence [I]

Five-stage, Ø150 mm Kühni; photographs at distributor stage 0 and active stages 1, 3 and **5**; Ø85 mm rotor, 70 mm spacing, 30% stator free area; 60–180 rpm, 1.24/2.00 L/min phase flows. Water continuous / Exxsol D-80 dispersed, mutually saturated, no mass transfer. This is a real terminal **active-compartment** DSD, not an outlet-flux or quiet-zone measurement [S1].

The fitted lognormal is **number weighted**. Its width s is the standard deviation of $\ln(d)$, not variance. At least 400 photographed drops per condition; 18 classes of 0.5 mm, observed sizes 0.5–8.5 mm. No validated sub-500- μ m completeness limit was established. A lowest observed size is not a physical lower cutoff.

Donor widths [I]: nominal regression $s = 0.2992 - 0.373984$ within the experiment; graphical fitted widths approximately 0.26–0.44 across all conditions. The latter is neither a terminal-only interval nor an RRBO confidence band. Regression $R^2 = 0.67$; roughly 20% width agreement in the donor study.

Transfer comparison [I unless C]	Donor S1	Fixed target
Continuous viscosity, Pa·s	0.0011	0.0598 (54.4× [C])
μ dispersed / μ continuous	≈1.45	0.02368 [C]
Rotor Reynolds number	7225–21675	≈388 [C], about 19× below donor minimum
Dispersed-phase buoyancy	Light drops rise	Heavy NMP-rich drops normally descend
Rotor / column diameter	0.567 [C]	0.330 [C]
Stage height / column diameter	0.467 [C]	0.300 [C]

Matching a mean, rpm or assigned power number cannot repair the rheology, Reynolds number, geometry, mass-transfer and direction differences. Do not extrapolate the donor width correlation to this service. Even “final stage” refers to a different dispersed-phase trajectory.

Other primary evidence: what is and is not transferred

S2 Kentish (1997), abstract only: static photographic and dynamic probe distributions differ through transport. **S3 Kentish (1998), abstract only:** axial DSD profiles were used to fit breakage/coalescence closure; no universal rate follows. **S4 Asadollahzadeh (2017), abstract/previews:** genuine Kühni tests compare maximum entropy, Gamma, inverse Gaussian and Weibull; maximum entropy reportedly fit best. No accessible target-transferable shape parameters.

Thus lognormal is an evidence-informed comparison, not a universal family. Distributor volume distributions in Garthe and breakup daughter kernels are not terminal ensemble DSDs; Shirvani mean-size correlations and Gomes abstract-level local DSD evidence do not fill the outlet-tail gap [S5–S7].

One mean does not identify the fine-volume population

Inherited premise: d32 = 6.068 mm is calculated for the active population, not measured at the terminal exit. All results on this page are [C], conditional on *the same population* having that mean [A]. Spherical or consistently volume-equivalent diameter weighting is assumed.

$$M_k = E_N[d^k]; \quad d32 = M_3 / M_2$$
$$dP_V = d^3 dP_N / M_3; \quad E_V[1/d] = 1/d32$$
$$F_V(c) \leq c E_V[1/d] = c/d32 \quad (0 < c < d32)$$

d32 is a volume-weighted harmonic mean, not the number median, minimum or a percentile. These normalization and inverse-moment constraints do not fix the CDF.

Cutoff [E], mm	Lower fraction [C], vol%	Upper supremum [C], vol%
0.500	0	8.24
1.000	0	16.48
1.455	0	23.98
2.081	0	34.29

Not outlet bounds. These bounds apply only to that same local population. They cannot constrain a selectively transported outlet distribution by assigning it the compartment d32. No positive minimum fine fraction follows.

The upper values are loose **suprema**: with finite positive drops and $c < d32$, the fraction is strictly below $c/d32$. Approaching the supremum requires increasingly large coarse drops and a fine atom approaching c . They are not probable tails or an implied physically realizable giant-drop population. Strict “below” and “at or below” agree for continuous distributions; cutoff atoms require care.

One constructive counterexample

Assume [A] two diameters, **0.4 and 12 mm**. Set small-drop volume share $w = (1/d32 - 1/12)/(1/0.4 - 1/12)$. Then [C] **3.371%** of volume is below *every* requested cutoff, with **99.894%** of the count in small drops. The remaining volume at 12 mm reconstructs $d32 = 6.068$ mm.

A monodisperse population at 6.068 mm [A] instead has zero below all four cutoffs [C], with exactly the same mean. These are mathematical counterexamples, not proposed physical DSDs. A fine satellite mode can escape a single-lognormal assumption.

Independent verification against the moment report

The new generator reconstructs these bounds and the bimodal mean, and independently compares the earlier GSD = 1.3, 1.5, 2.0 and 2.5 cases. Number moments $k = 0-3$, d32, volume inverse moment and all four truncated volume fractions agree within numerical tolerances. **PASS — mathematical consistency only.**

Conditional tails — all four requested cutoffs

Assume a number-lognormal population [A], $\ln(d/1\text{ mm}) \sim \text{Normal}(m,s^2)$, anchored to inherited d32 = 6.068 mm. Rescaling a donor width does **not** predict the RRBO/NMP tail or produce a confidence band.

$$M_k = \exp(km + k^2s^2/2) \text{ mm}^k; \quad m = \ln(6.068) - 2.5s^2$$
$$F_N(c) = \Phi([\ln(c/d32) + 2.5s^2]/s)$$
$$F_V(c) = \Phi([\ln(c/d32) - 0.5s^2]/s)$$

Dispersed-volume fraction below cutoff, % [C | A]

s [A]	0.500 mm	1.000 mm	1.455 mm	2.081 mm
0.35	1.37×10^{-11}	5.01×10^{-6}	0.00104	0.0613
0.44	1.89×10^{-7}	0.000788	0.0265	0.400
0.70	0.00450	0.172	0.842	3.01
1.00	0.137	1.06	2.69	5.82

s = 0.35 and 0.44: conditional width transfers inside the approximate donor graphical envelope; neither a fitted target width nor a probability band. **s = 0.70 and 1.00:** deliberately broad analyst stress assumptions, not literature recommendations. All four columns are reporting cutoffs [E], not selected capture duties.

Number fraction below the same cutoff, % [C | A]

s [A]	0.500 mm	1.000 mm	1.455 mm	2.081 mm
0.35	1.96×10^{-8}	0.000949	0.0675	1.45
0.44	0.000240	0.136	1.60	9.14
0.70	3.47	20.4	38.6	58.8
1.00	50.2	75.7	85.8	92.4

At s = 0.44, below 2.081 mm: **9.14% of count** but **0.400% of dispersed volume** [C | A].
Neither quantity is the raffinate-path flux fraction or NMP concentration.

Extremely small numbers are retained in scientific notation, not rounded to a misleading zero. Their displayed precision is computational, **not empirical resolving power**: the donor photographs cannot establish such trace tails. Unknown detection bias, feed-generated fines or a second mode may dominate these mathematical extrapolations.

Numerics: the smaller normal tail is integrated directly by composite Simpson quadrature, factoring out $\exp(-z^2/2)$; no subtraction of nearly equal numbers. Reference CDF checks include $z = 0, -1, -3$ and -8 ; doubling 16,000 panels checks stability. Independent normal-coordinate integration verifies moments and truncated third moments. These are calculation checks, not physical validation.

Transport and coalescence: the missing link to carryover

Quantity	What it represents / missing evidence
Number DSD	Local counts of individual drops; image detection and sampling bias matter.
Local volume DSD	Counts weighted by d^3 ; specifies relative local dispersed inventory, not load.
One-way upward volume-flux DSD	Crossing volume weighted by outward velocity over the actual exit plane; not a snapshot.
Quiet-region / outlet DSD	Evolves through selective return, coalescence, residual breakup and feed interaction.
Physical carryover	Dispersed NMP-rich droplets in the withdrawal; distinct from molecularly dissolved NMP.

In an ideal dilute uniform counterflow [A], outward volume flux is proportional to $d^3 n(d) \max[U - vt(d), 0]$. Where settling speed grows with size, this preferentially transports fines. The normalized outlet tail can be much larger than the local-volume tail. Real stator jets, recirculation, turbulent exchange and nonuniform holdup prevent treating this illustration as a calibrated model [S2]. Upward and downward crossings must be resolved separately.

Fresh NMP introduced above the terminal stator is another possible droplet source.

Measuring only the final active compartment misses feed jets, distributor breakup and possible short-circuit paths to withdrawal. Heavy NMP normally travels downward; upward escaped drops are a selected population.

Coalescence is plausible, not quantified

After the final rotor/stator, reduced breakup may permit growth. But drops must collide, drain the continuous-phase film and merge. Collision frequency depends on concentration, size, differential settling and shear; merger efficiency depends on composition, contaminants, interfacial mobility and contact duration. Quiet conditions may reduce both breakup *and collisions*. Sparse fines can persist.

S3 fitted rates to axial DSDs and transport; it provides no universal “growth per second,” no service-specific coalescence time and no justified growth credit here. A rising mean alone is not proof of merger: selective departure changes the measured population. Batch settling and material-specific coalescing-aid studies [S8, S9] suggest methods, not target parameters.

No actual ppm, kg/h, holdup or entrainment load is calculated. Relative tails cannot supply total escaped dispersed flow, composition or permitted residual. Raffinate flow is not entrained solvent throughput.

If entering physical dispersed flow and its flux DSD were measured, a no-growth grade-efficiency balance could integrate residual load. With coalescence, capture also depends on load, residence paths and accumulated inventory. None is closed here. Droplet volume is not automatically pure-NMP mass; dissolved NMP is not gravity-removable.

The 1600 mm reservation [I/E] is not entirely quiet liquid. The prior 600 mm clear allowance [I/E] corresponds to a nominal $L/U \approx 218 \text{ s}$ [C], **not a coalescence time or validated residence distribution**. No minimum quiet height is derived.

Compare concepts without preselecting equipment

Frozen basis [I]: Ø700 ×4200 mm active section; 20 ×210 mm compartments; Ø231 mm rotor; 30 rpm; assigned $N_p = 1.2$; stator free area 0.40. Entire bottom unchanged. At 40 °C: $\rho_c/\rho_d = 869/1015 \text{ kg/m}^3$, $\mu_c/\mu_d = 0.0598/0.001416 \text{ Pa}\cdot\text{s}$, interfacial tension 0.011 N/m; raffinate 3.8218 m³/h. These are inherited, not newly validated outlet properties.

Conditional local-area sensitivity — not approved enlargement

ID, mm [I/E]	Area m ² [C]	U mm/s [C]	vt = U, mm [C]	vt = 2U, mm [C]
700 [I]	0.385	2.759	1.455	2.081
900 [E]	0.636	1.669	1.127	1.604
1000 [E]	0.785	1.352	1.013	1.440
1200 [E]	1.131	0.939	0.843	1.196

Schiller–Naumann immobile spherical isolated-drop model [A]: $Re = \rho_c vt d/\mu_c$; $CD = (24/Re)(1+0.15Re^{0.687})$; $CD \rho_c vt^2/2 = (2/3)d \Delta\rho g$. The vt = 2U column retains the former engineer-selected $f = 0.5$ margin [E], **not a grade-efficiency law**. Rounded baseline values explain 1.455 and 2.081 mm; they are velocity equalities, not percentiles.

Candidate	Present disposition	Evidence that could justify it
Bare Ø700 ×1600 reservation	Baseline candidate; performance HOLD, not PASS	Actual physical carryover meets specification with stable top inventory and return across representative conditions.
Modest area increase	No change justified now; comparison only	Measured limiting transported load lies in a size range materially helped by lower U; validate flow and full return path.
Internal collector / aid	Not mandated by a 500 µm failure	Representative capture/coalescence, compatibility, drainage, fouling, pressure loss and access evidence.
External guard	Not mandated or preselected	Actual withdrawal load and specification justify independent separation; demonstrate collection, pressure/control and return.

The unchanged Ø700 return throat remains controlling

An enlarged region may allow local settling but cannot make an unchanged drop descend through the faster upflow below. For example, a 1 mm drop has $vt \approx 1.317 \text{ mm/s}$ [C | A]: it can settle in the 1200 mm comparison, but not through Ø700. Enlargement alone is not an integrated return solution.

Growth, a demonstrated descending collected phase/film, or a protected return route needs evidence. A drain requires a real collector, pressure/elevation balance, losses, liquid seal and capacity at the *entrained* load. Do not assume a top heavy-phase interface, wall film or spare volume. **No new geometry, giant vessel or equipment purchase is recommended.**

Qualification and reproducible evidence record

Next scientific qualification — actual duty first

At representative **40 °C and 30 rpm [I]**, use actual equilibrated/aged RRBO-NMP, phase rates, water/solids/contamination history and internals. Obtain matched raw drop sizes or $n(d)$ in the **last active compartment, immediately above the terminal stator, above fresh-NMP feed, along the available quiet region, and actual raffinate withdrawal**. Record axial/radial position, time, spatial versus one-way-flux weighting and withdrawal bias.

Validate optical or minimally disruptive sampling for viscous/opaque material: report resolution and calibrated recovery below all four cutoffs, unresolved sub-resolution volume, sample count, uncertainty and induced breakup/coalescence. Report measured d_{32} , number/volume CDFs, $dV_{10}/dV_{50}/dV_{90}$ and F_v at **0.500, 1.000, 1.455 and 2.081 mm [E]**; test a second fine mode rather than force a lognormal. Separately quantify actual physical dispersed carryover, dissolved NMP, droplet composition and a matched mass balance.

Use time-resolved DSD/merger observations and representative-material settling data to test post-stator kinetics and stable return. Define permitted physical carryover and relevant normal, startup and surge conditions before qualification. **No arbitrary duration or sample count is declared sufficient**. Precision and operating coverage must discriminate against the agreed specification. Qualify the bare candidate first; evaluate aids or area only if evidence identifies the limitation.

Primary references and access limits

Access below is as documented in the supplied research reviews; no new online retrieval. No retrieved source closes the full terminal-stator-to-outlet RRBO/NMP chain.

1. **S1 Oliveira et al. (2008), terminal DSD.** [doi:10.1590/S0104-66322008000400010](https://doi.org/10.1590/S0104-66322008000400010). Full HTML/PDF text and equation images inspected in source review; pp. 731–736, Table 1, Figs. 2–5, Eqs. 2–5. Direct PDF curl gave HTML; not misrepresented as downloaded PDF.
2. **S2 Kentish et al. (1997), drop velocities.** [doi:10.1021/ie9702690](https://doi.org/10.1021/ie9702690). Publisher abstract only; full method and envelope unavailable.
3. **S3 Kentish et al. (1998), coalescence/breakage rates.** [doi:10.1021/ie970336q](https://doi.org/10.1021/ie970336q). Publisher abstract only; no fitted rates transferred.
4. **S4 Asadollahzadeh et al. (2017), DSD families.** [doi:10.1016/j.cherd.2016.08.025](https://doi.org/10.1016/j.cherd.2016.08.025). Abstract/section previews; no full parameter tables.
5. **S5 Garthe (2006), dissertation.** <https://mediatum.ub.tum.de/601973>. Repository/full existing text; initial q3 distributions are not terminal tails.
6. **S6 Shirvani et al. (2016).** [doi:10.5829/idosi.ijs.2016.29.03c.02](https://doi.org/10.5829/idosi.ijs.2016.29.03c.02). Full existing primary text; mean-size/flooding, not verified outlet width.
7. **S7 Gomes et al. (2009).** [doi:10.1021/ie801034a](https://doi.org/10.1021/ie801034a). Publisher abstract only; local DSD/mass-transfer effects, no transferred parameters.
8. **S8 Henschke et al. (2002).** [doi:10.1016/S1385-8947\(01\)00251-0](https://doi.org/10.1016/S1385-8947(01)00251-0). Abstract/introduction/previews; batch coalescence characterization.
9. **S9 Schäfer et al. (2022).** [doi:10.1002/cjce.24503](https://doi.org/10.1002/cjce.24503). Open full text, relevant sections reviewed; material-specific coalescing aids, not NMP qualification.

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